Wetland_fraction_r2_WIP.R

August 9, 2024

Contents

`NWI_Wetland_fraction` *Create gridded wetland coverage maps using National Wetlands Inventory data*

Description

'NWI_Wetland_fraction' writes multiple tiff files of the fractional wetland cover in each pixel. There is one file for each state - wetland type combination.

Usage

```
NWI_Wetland_fraction(
  input_directory,
  output_directory,
  domain,
  state_name_list
)
```

Arguments

`input_directory`
: Character providing the full filepath to save/load input data

`output_directory`
: Character providing the full filepath to save processed data

`domain`
: SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system

`state_name_list`
: Character vector listing all states within the desired domain

Details

This function downloads National Wetlands Inventory (NWI) data and calculates the fraction of each gridcell in the domain that is covered by the NWI polygons. This is done separately for each state as the data is available by state or watershed. It also processes each type of wetland separately.

The wetland types are pulled from the NWI system, subsystem, and class as defined here https://www.fws.gov/media/national-wetland-inventory-wetlands-and-deepwater-map-code-diagram and discussed in detail here https://www.fws.gov/media/classification-wetlands-and-deepwater-habitats-u They are M1 (marine, subtidal), M2 (marine, intertidal), E1 (estuarine, subtidal), E2 (estuarine, intertidal), R1 (riverine, tidal), R2 (riverine, lower perrennial), R3 (riverine, upper perennial), R4 (riverine, intermittent), L1 (lacustrine, limnetic), L2 (lacustrine, littoral), PFO (palustrine, forested), and PNF (palustrine, all non-forested classes).

The NWI data for each state will be automatically downloaded. This can be up to several GB per state.

The NWI is a biannually updated dataset including the extent and type of wetlands across the United States. It is available at https://www.fws.gov/program/national-wetlands-inventory/wetlands-data.

A separate TIFF file is saved for each state - wetland type combination. These are used in further processing.

Value

Nothing is returned from the function, but the main outputs are TIFF files of the fractional wetland coverage per pixel for each wetland type and state. They are titled "state abbreviation _ wetland type abbreviation.tiff".

Author(s)

Joe Pitt, <madeup@wisc.edu>

Kris Hajny, <blank@fake.edu>

Israel Lopez-Coto, <test@test.edu>

Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
            ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
            xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
            crs=grid_crs)

NWI_Wetland_fraction(input_directory="~/../Desktop/input/",
                     output_directory="~/../Desktop/",
                     domain=grid,
                     state_name_list=c("DE","MD","NJ","NY","PA"))
```

Index